

AI-adopting firms have so far benefited from higher wages. The productivity benefits are unevenly distributed, with medium and larger firms experiencing significantly higher productivity gains than their smaller peers. The stratification highlights concerns that AI could exacerbate income gaps (Cazzaniga et al., 2024; Cornelli, Frost, and Mishra, 2023), particularly given Europe’s SME-dominated industrial structure.

The findings carry significant implications for European policy makers seeking to harness the productivity-enhancing potential of AI while ensuring inclusive and balanced growth. Given the disruptive nature of AI and the higher returns associated with AI adoption for larger companies, a balanced policy approach should focus on creating an environment suitable for small and innovative firms’ growth, accompanied by targeted incentives to fully exploit the benefits of AI adoption. In this context, developed financial markets play a crucial role.

Our results also highlight the importance of complementary investments in intangible assets – particularly software and data or workforce training – in realizing AI’s productivity dividends. To maximize the returns from AI deployment, public policy must go beyond hardware subsidies and incentivize firm-level investment in integration, workflow redesign, and continuous learning. Workforce re-skilling programs should prioritize “fusion skills”, such as prompt engineering, data stewardship and human-in-the-loop decision-making that enhance human-AI complementarity.

The structure of the paper is as follows. Section 2 reviews the related literature on AI adoption, productivity and labor market effects. Section 3 describes the data and presents stylized facts that motivate the empirical strategy. Section 4 outlines our identification approach and empirical methodology. Section 5 reports the main results on productivity and employment, along with several robustness checks. Section 6 concludes by discussing policy implications and avenues for future research.

2 Related literature

Our study engages with three interconnected strands of research on AI and its economic implications. First, the literature on firm-level technology adoption provides critical frameworks for understanding AI diffusion patterns. Seminal studies by Comin and Hobijn (2010) and Comin and Mestieri (2018) establish that technological innovations

diffuse unevenly across countries, with adoption lags determined by human capital and predecessor technologies – a pattern we observe in Europe’s AI uptake. Recent studies by [Babina et al. \(2024\)](#) and [Bonney et al. \(2024\)](#) document robust AI adoption among US firms, particularly large innovators in knowledge-intensive sectors. Yet evidence for non-US firms remains sparse, with [Rückert, Weiss, and Revoltella \(2020\)](#) being a notable exception that documents Europe’s adoption gap with the US.

Our paper extends this work by demonstrating how capital intensity, the financial dependence theory of [Rajan and Zingales \(1998\)](#) and firm size mediate AI adoption in Europe – findings that align with the analysis of technology-job complementarities of [Acemoglu et al. \(2022\)](#). This seems particularly relevant in the context of the growing importance of AI adoption for EU labor markets. Using occupation-level data for 16 European countries over 2011-2019, [Albanesi et al. \(2024\)](#) document that employment shares increase in occupations more exposed to AI, with the strongest gains in occupations employing younger and higher-skilled workers.

Second, research on the productivity effects of AI highlights stark contrasts between macroeconomic projections and microeconomic evidence. Most macro studies suggest relatively modest aggregate effects: for example, [Acemoglu \(2024\)](#) estimates only 0.07% US annual growth in total factor productivity over a decade, while the general equilibrium model of [Filippucci, Gal, and Schief \(2024\)](#) shows how Baumol’s disease effects could limit gains to 0.2–0.6 percentage points annually. In contrast, micro-level field experiments report large efficiency improvements in the short run: [Noy and Zhang \(2023\)](#) and [Brynjolfsson, Li, and Raymond \(2025\)](#) find 14–40% productivity gains in writing and customer service tasks, with [Gambacorta et al. \(2024\)](#) showing similar effects for coding. Our results reconcile this divergence: we identify a causal 4% labor productivity gain at the firm level, echoing the mid-range estimates of [Bergeaud \(2024\)](#), while confirming that micro efficiency improvements are not fully reflected in aggregate growth due to adoption frictions.

Third, the literature on distributional effects underscores the unequal impact of AI across workers, firms, and regions. [Cazzaniga et al. \(2024\)](#) predicts widening income inequality due to AI-skilled labor complementarity, while [Hennig and Khan \(2025\)](#) warns of cross-country divergence as AI benefits concentrate in advanced economies. The concerns resonate with evidence that technology diffusion gaps explain at least 25% of global income differences ([Comin and Mestieri, 2018](#)). Our heterogeneity analysis validates these

predictions, showing larger productivity gains in medium and large firms compared to micro and small enterprises.

Our contribution bridges these strands by: (i) providing the first causal evidence on the effects of AI adoption on firm productivity and employment effects in Europe, addressing gaps noted by [Aldasoro et al. 2024](#), (ii) developing a novel IV strategy that extends [Rajan and Zingales \(1998\)](#)’s financial dependence approach to technology diffusion, (iii) presenting evidence of capital-deepening, rather than labor-replacing, effects that can contribute to refining [Aghion and Bunel \(2024\)](#)’s growth theories, and (iv) discussing which investment strategies enable the productivity dividends of AI.

3 Data

We use data from the European Investment Bank Investment Survey (EIBIS), an annual survey initiated in 2016 encompassing approximately 12,000 non-financial corporations across the EU. This survey employs a stratified random sampling method, targeting firms in all 27 EU countries.² Since 2019, EIBIS also covers a sample of 800 firms in the US, allowing for a comparison of investment dynamics between the EU and the US. The respondents to the interviews are senior managers or financial directors with responsibility for investment decisions and how investments are financed – for example, the owner, chief financial officer or chief executive officer. [Brutscher et al. \(2020\)](#) provide evidence that EIBIS is representative of the business population in the EU, as described by Eurostat’s Structural Business Statistics.

Since 2019, firms participating in EIBIS are asked about the use of big data analytics and AI. In the EIBIS questionnaire, big data analytics and AI are defined as technologies that intelligently automate tasks and provide insights that augment human decision making, like machine learning, robotic process automation, natural language processing (NLP), algorithms, neural networks. Firms are asked the following question: “To what extent, if at all, are big data analytics and artificial intelligence used within your busi-

²The EIBIS sample is stratified disproportionally by country, sector and firm size class, and proportionally by region within each country. The firms have at least five employees, with both full-time and part-time employees being counted as one employee, and employees working less than 12 hours per week being excluded. An enterprise is defined as a company trading as its own legal entity. As such, branches are excluded from the target population. However, the definition is broader than in a typical enterprise survey because some company subsidiaries are their own legal entities.

ness? A. Not used in the business. B. Used in parts of the business. C. Entire business is organized around this technology.” We define AI adopters as firms that use big data analytics and AI in parts of the business or if the entire business is organized around AI.

EIBIS provides a comprehensive dataset that captures a wide spectrum of firm-level characteristics and investment-related information across surveyed firms. It includes quantitative data – such as annual turnover, number of employees, value of fixed assets, total wages, investment in different categories of tangible and intangible assets – as well as demographic data, for instance on firm age and sector, offering a deep understanding of the structural profile of businesses. Beyond these metrics, EIBIS also gathers qualitative insights through survey responses, shedding light on firms’ internal perceptions and strategic outlooks. This includes assessments of managerial competencies, financing strategies, perceived barriers to invest and investment focus.

To enrich and validate the self-reported survey data, EIBIS is complemented by administrative financial information sourced from Moody’s ORBIS database. This integration allows for a more robust analysis by incorporating detailed financial statements on balance sheets and profit and loss accounts. The combination of subjective perceptions and objective financial data enables a better understanding of investment dynamics and firm performance in different sectors, countries and market segments.

3.1 Descriptive statistics and stylized facts

AI innovation in Europe continues to lag behind the United States. In particular, the EU trails the US in the number of AI-related patents, with the gap widening further after the pandemic (EIB, 2025). While the absolute number of AI patents is higher in the US than in the EU (Figure 1a), the higher share of AI patents relative to total patents also underscores the stronger specialization in AI innovation in the US (Figure 1b).